

Sebastiano Gaiardelli

Via Don Bosco, 10 Affi(VR), Italy

+39 349 340 2945

sebastiano.gaiardelli@univr.it

linkedin.com/in/sebastianogaiardelli

Born 16 December 1997

EDUCATION

2021-2024

PhD student in Computer Science and Engineering

University of Verona

- Industry 4.0
- Cyber-physical production systems (CPPS)
- Production scheduling
- Knowledge representation
- Manufacturing Execution System (MES)

2019-2021

Master's degree in Computer Science and Engineering

University of Verona

- Final grade of 110/110 cum laude
- Supervisor: Prof. Franco Fummi
- Title of the thesis: "Meta-MES: A Scheduling Architecture for Flexible Automation"

2016-2019

Bachelor's Degree in Computer Science

University of Verona

- Final grade of 110/110
- Supervisor: Dr. Bogdan Mihai Maris
- Title of the thesis: "Continuous representation of digital signal"

WORK EXPERIENCE

September 2023 – December 2023

Contract for Occasional Work

University of Verona

- Developed an optimization component for the production recipes executed by Adaptive Manufacturing Controller (AMC) in the ICE Laboratory.

October 2022 – November 2022

Contract for Occasional Work

University of Verona

- Focused on the integration of an ATE machine with the production recipes
- Development of a specific interface within the Meta-MES architecture to support ATE machines.

August 2021 – September 2021

Contract for Occasional Work

University of Verona

- Research activity in the context of Industry 4.0
- Development of a Meta-MES architecture to control CPPS.

December 2019 – July 2020

Software Engineer

Dovit Italia S.r.l.

- Focused on the integration of new device driver and new functionalities for the IoT gateway "Do.Control"
- Developed and managed the cloud infrastructure managing the integration between the IoT gateway with many cloud services (e.g., Google Home, Alexa, IFTTT, Spotify, Telegram)
- Owned technical leadership responsibilities for research and development of future product features and services

December 2018 – December 2019

Software Engineer

Home Innovation S.r.l.

- Focused on the integration of new device driver and new functionalities for the IoT gateway "Do.Control"
- Developed and managed the cloud infrastructure managing the integration between the IoT gateway with many cloud services (e.g., Google Home, Alexa, IFTTT, Spotify, Telegram)
- Owned technical leadership responsibilities for research and development of future product features and services

TEACHING EXPERIENCE

October 2023 – July 2024

Teaching Assistant (14 hours)

- Course Name: Embedded & IoT System Design
- Master Degree: University of Verona, Computer Engineering for Robotics and Smart Industry
- Language: English

October 2022 – July 2023

Teaching Assistant (14 hours)

- Course Name: Embedded & IoT System Design
- Master Degree: University of Verona, Computer Engineering for Robotics and Smart Industry
- Language: English

October 2022 – July 2023

Teaching Assistant (30 hours)

- Course Name: Architettura degli elaboratori
- Degree: University of Verona, Computer Science
- Language: Italian

October 2021 – July 2022

Teaching Assistant (30 hours)

- Course Name: Architettura degli elaboratori
- Degree: University of Verona, Computer Science
- Language: Italian

VISITING EXPERIENCE

February 2024 – August 2024

Technische Universität München (TUM) - Institute of Automation and Information Systems

- Visiting Ph.D. student
- Supervisor: Prof. Dr.-Ing. Birgit Vogel-Heuser

PUBBLICATIONS

[1] S. Gaiardelli, J. Wilch, F. Fummi and B. Vogel-Heuser "Automating CPPSs Analyses with SysML v2 and Lingua Franca in the Context of Industry 4.0," (IECON 2024) - 50th Annual Conference on IEEE Industrial Electronics, 2024.

[2] P. Turco, E. Zanella, A. Valentini, S. Gaiardelli, N. Dall'Ora, M. Lora and F. Fummi "Digital Twin Integration using Lingua Franca and FMI for Testing Factory Automation Software," (IECON 2024) - 50th Annual Conference on IEEE Industrial Electronics, 2024.

[3] S. Gaiardelli, D. Carra, S. Spellini and F. Fummi, "Dynamic Job and Conveyor-based Transport Joint Scheduling in Flexible Manufacturing Systems," Applied Sciences, 2024.

[4] S. Gaiardelli, S. Spellini, M. Panato, C. Tadiello, M. Lora, D. S. Cheng and F. Fummi, "Enabling Service-oriented Manufacturing through Architectures, Models and Protocols," IEEE Access, vol. 12, pp. 85259-85274, 2024, doi: 10.1109/ACCESS.2024.3385634.

[5] S. Gaiardelli, M. Lora, S. Spellini and F. Fummi, "RRPDG: A Graph Model to Enable AI-Based Production Reconfiguration and Optimization," in IEEE Transactions on Industrial Informatics, doi: 10.1109/TII.2024.3352645.

[6] M. Libro, S. Gaiardelli, M. Lora, F. Fummi, "Integrating Modeling Languages with Ontologies in the Context of Industry 4.0," 2024 IEEE International Conference on Industrial Technology (ICIT), 2024, pp. 1-7, doi: 10.1109/ICIT58233.2024.10540801.

[7] S. Gaiardelli, N. Dall'Ora, F. Ponzio, E. Fraccaroli, F. Fummi, S. Di Cataldo et al., "A Data Fusion Service-Oriented Infrastructure for Production Line Monitoring," 2024 IEEE International Conference on Industrial Technology (ICIT), 2024, pp. 1-8, doi: 10.1109/ICIT58233.2024.10541026.

[8] A. Mascolini, S. Gaiardelli, F. Ponzio, N. Dall'Ora, E. Macii, S. Vinco et al., "VARADE: a Variational-based AutoRegressive model for Anomaly Detection on the Edge," 2024 61st ACM/IEEE Design Automation Conference (DAC), 2024.

[9] M. Lora, S. Gaiardelli, C. Oh, S. Spellini, P. Nuzzo, and F. Fummi, "Design Automation for Cyber-Physical Production Systems: Lessons Learned from the DeFacto Project," 2024 Design, Automation & Test in Europe Conference & Exhibition (DATE) , 2024, pp. 1-6, doi: 10.23919/DATE58400.2024.10546563..

[10] K. Alamin, D. Appello, A. Beghi, N. Dall'Ora, F. Depaoli, S. Di Cataldo et al., "An AI-Enabled Framework for Smart Semiconductor Manufacturing," 2024 Design, Automation & Test in Europe Conference & Exhibition (DATE) , 2024, pp. 1-6, doi: 10.23919/DATE58400.2024.10546768.

[11] L. Capogrosso, A. Mascolini, F. Girella, G. Skenderi, S. Gaiardelli, N. Dall'Ora et al., "Neuro-Symbolic Empowered Denoising Diffusion Probabilistic Models for Real-Time Anomaly Detection in Industry 4.0: Wild-and-Crazy-Idea Paper," 2023 Forum on Specification & Design Languages (FDL), Turin, Italy, 2023, pp. 1-4, doi: 10.1109/FDL59689.2023.10272095.

[12] A. Mascolini, S. Gaiardelli, F. Ponzio, N. Dall'Ora, E. Macii et al., "Robotic Arm Dataset (RoAD): a Dataset to Support the Design and Test of Machine Learning-driven Anomaly Detection in a Production Line," (IECON 2023) - 49nd Annual Conference on IEEE Industrial Electronics, 2023, pp. 1-7, doi: 10.1109/IECON51785.2023.10311726.

[13] S. Gaiardelli, S. Spellini, M. Pasqua, M. Ceccato and F. Fummi, "Integrating Smart Contracts in Manufacturing for Automated Assessment of Production Quality," (IECON 2022) - 48nd Annual Conference on IEEE Industrial Electronics, 2022, pp. 1-6, doi: 10.1109/IECON49645.2022.9968887.

[14] K. Alamin, Y. Chen, S. Gaiardelli, S. Spellini, A. Calimera et al., "SMART-IC: Smart Monitoring and Production Optimization for Zero-waste Semiconductor Manufacturing," 2022 23th Latin-American Test Symposium (LATS), 2022, pp. 1-6, doi: 10.1109/LATS57337.2022.9937011.

[15] S. Gaiardelli, D. Carra, S. Spellini and F. Fummi, "On the Impact of Transport Times in Flexible Job Shop Scheduling Problems" 2022 27th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA), pp. 1-8, doi: 10.1109/ETFA52439.2022.9921441.

[16] S. Gaiardelli, S. Spellini, M. Panato, M. Lora and F. Fummi, "A Software Architecture to Control Service-Oriented Manufacturing Systems," 2022 Design, Automation & Test in Europe Conference & Exhibition (DATE), 2022, pp. 40-43, doi: 10.23919/DATE54114.2022.9774522.

[17] S. Gaiardelli, S. Spellini, M. Lora and F. Fummi, "A Hierarchical Modeling Approach to Improve Scheduling of Manufacturing Processes," 2022 IEEE 31st International Symposium on Industrial Electronics (ISIE), 2022, pp. 226-232, doi: 10.1109/ISIE51582.2022.9831468.

[18] S. Gaiardelli, S. Spellini, M. Lora and F. Fummi, "Modeling in Industry 5.0: What Is There and What Is Missing: Special Session 1: Languages for Industry 5.0," 2021 Forum on specification & Design Languages (FDL), 2021, pp. 01-08, doi: 10.1109/FDL53530.2021.9568371.

[19] S. Spellini, S. Gaiardelli, M. Lora and F. Fummi, ”Enabling Component Reuse in Model-based System Engineering of Cyber-Physical Production Systems,” 2021 26th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA), 2021, pp. 1-8, doi: 10.1109/ETFA45728.2021.9613572.

PERSONAL SKILLS AND COMPETENCES

Languages English: Intermediate written and spoken.
Italian: Native speaker.

Personal skills Motivated | Organized | Responsible | Ambitious | Energetic | Independent | Versatile | Decision-making | Cooperative | Team leadership | Goal setting | Determined | Accurate

Digital Skills Linux (Debian/Ubuntu/RedHat) Distributions | Windows | Microsoft Office | Libre Office | Git | Object-Oriented Programming | Database | Python | C# | C++ | Java | C | Javascript | HDL Languages | LaTeX | HTML | BASH | Edge Computing | Embedded Software | Cloud computing | AWS | Google Cloud Platform | Kubernetes | Docker | Virtualization | OPC UA | IoT | IIoT | CPPS | CPS

I authorize the handling of my personal data according to the Personal Data Protection Code - Legislative Decree n. 196/03. and GDPR 679/16 ”European regulation on the protection of personal data”.

Verona 27th August 2024,